From Complaints to Action

PromptWars x ASTRA | AI for Civic Impact | Problem Statement

Format	Individual / solo	**Time limit**	8 hours	**Output**	Working prototype + demo

1. Background

Every day, communities face waterlogging, damaged roads, overflowing waste bins, broken streetlights and disrupted public services. Reports reach authorities through many channels: calls, messages, social media, walk-ins. The same issue is often reported ten times, urgent hazards sit in the same queue as routine requests, and decision-makers cannot tell what needs attention first. Citizens, meanwhile, rarely see what happened to their complaint.

2. The Mission

> *Build an AI-powered platform that transforms scattered citizen reports into prioritized, explainable and trackable action.*

A system that only collects complaints is not enough. Yours should **understand** reports, **detect patterns**, **surface hotspots**, and help a decision-maker choose **what to fix first, and why**.

3. Core Requirements

Your prototype must cover all four stages of the report lifecycle.

Stage	What the platform must do
1. Capture	Let users submit a problem as text and/or an image, choose a category, and give a location (map pin, address or coordinates). Reporting should take under a minute and work on a phone-sized screen.
2. Understand	Use AI to categorize the report, summarize the description, flag likely duplicates, and assign a priority based on urgency and potential impact.
3. Reveal	Show reports on a map or dashboard. Make recurring problems, high-priority locations and area-level patterns visible, especially ones that are hard to spot by reading a list.
4. Act	Provide a workflow to review, assign and track each report from submission to resolution, with visible status so that submitters can follow progress.

4. The AI Requirement

Your platform must demonstrate **at least one meaningful AI capability beyond a basic chatbot**. Choose what you can build well and explain clearly. Examples:

- **Duplicate detection:** identify multiple reports describing the same underlying issue and merge or link them.
- **Explainable prioritization:** show why one report outranks another, in plain language, with the factors used.
- **Cluster / hotspot detection:** detect a group of related problems in an area (for example, drain blockage, waterlogging and road damage on one street).
- **Action briefs:** convert unstructured reports into a concise brief a decision-maker can act on in under a minute.

You may use any model, API, library or rule-based method, including classical ML, embeddings, LLMs or hybrids. What matters is that the AI does real work in the pipeline and that you can explain how it works.

5. Final Demonstration

At the end of eight hours, demonstrate one complete user journey (target 5 minutes):

- **Submit** a new report (text and/or image, with category and location).
- **Show how AI processes it:** category, summary, duplicate check, priority and the reasoning behind them.
- **Surface an insight:** a hotspot, cluster, trend or action brief that emerges from the data.
- **Track the action:** assign the report, move it through the workflow, and show the status update from the citizen's side.

A polished interface alone will not be enough. Judges will reward solutions that work, make sensible decisions and are honest about their limits.

6. Data

Public, sample or synthetic data are all allowed. We recommend preparing a synthetic set of **at least 30 to 50 reports** across multiple categories, including deliberate duplicates, a few urgent safety issues and at least one geographic cluster, so that your AI features have something meaningful to find. Do not use real personal data.

7. Rules

- Solo participation only. All work must be done within the 8-hour window.
- AI coding assistants and AI tools are permitted and encouraged, but you must understand your implementation and be able to explain it to judges.
- **Integrity:** do not claim live integrations, real-world deployments or AI capabilities that your prototype does not actually have. Mocked components must be labelled as mocked.
- Technology stack, interface and implementation strategy are entirely your choice.

8. Submission Checklist

- A working, demonstrable prototype (hosted link or runnable locally).
- Source code repository with a README covering setup, architecture and the AI approach.
- A short demo (live or recorded) of the complete user journey.
- A **Limitations and Honesty** note: what is real, what is mocked, where the AI can be wrong, and what you would do next.

9. Suggested Time Plan (optional)

Hours	Focus
0 - 1	Scope the MVP, choose the AI capability, design the data model, prepare synthetic data
1 - 3.5	Report capture, storage, and the AI pipeline (categorize, summarize, duplicates, priority)
3.5 - 5.5	Map / dashboard, hotspot or cluster insight, assignment and status workflow
5.5 - 7	Citizen tracking view, explanations, edge cases, UX polish
7 - 8	Testing, README, limitations note, demo rehearsal